THE CONORMAL SEQUENCE OF AN IMMERSION OF ALGEBRAIC SPACES OVER A BASE SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1144-1155 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 1157-1177, the complete author proof (19 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Immersions of algebraic spaces, conormal sheaves, differentials, and etale localisation compatibility; the construction is retained.

Source reference. Original lemma lemma-etale-conormal: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 432–441. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-functoriality-differentials: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1062–1083. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-localize-differentials: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 981–996. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-differentials-relative-immersion: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1177–1177. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-i-star-equivalence: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1177–1177. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let $i : Z \to X$ be an immersion of algebraic spaces over B. There is a canonical exact sequence*

$$\mathcal{C}_{Z/X} \to i^*\Omega_{X/B} \to \Omega_{Z/B} \to 0$$

where the first arrow is induced by $d_{X/B}$ and the second arrow comes from Lemma lemma-functoriality-differentials.

Proof. This is the algebraic spaces version of Morphisms, Lemma morphisms-lemma-differentials-relative-immersion and will be a consequence of that lemma by étale localization, see Lemmas lemma-localize-differentials and lemma-etale-conormal. However, we should make sure we can define the first arrow globally. Hence we explain the meaning of "induced by $\mathrm{d}_{X/B}$" here. Namely, we may assume that i is a closed immersion after replacing X by an open subspace. Let $\mathcal{I} \subset \mathcal{O}_X$ be the quasi-coherent sheaf of ideals corresponding to $Z \subset X$. Then $\mathrm{d}_{X/S} : \mathcal{I} \to \Omega_{X/S}$ maps the subsheaf $\mathcal{I}^2 \subset \mathcal{I}$ to $\mathcal{I}\Omega_{X/S}$. Hence it induces a map $\mathcal{I}/\mathcal{I}^2 \to \Omega_{X/S}/\mathcal{I}\Omega_{X/S}$ which is $\mathcal{O}_X/\mathcal{I}$-linear. By Morphisms of Spaces, Lemma spaces-morphisms-lemma-i-star-equivalence this corresponds to a map $\mathcal{C}_{Z/X} \to i^*\Omega_{X/S}$ as desired. □